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MILITARY SPECIFICATION

NETWORK D CIRCUIT BOARD: 8623286

1. SCOPE

1.1 This specification covers one type of network D circuit board, part of the Test Set, Computer Logic Unit AN/GSM-70.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-M-46357	Manufacturing Processes and Inspection of Computer, Gun Direction: M18; General Requirements for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U. S. Army Munitions Command

F8623286	Network D Assembly
F8623287	Schematic, Electrical, Network D

FSC 1220

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PACKAGING DATA SHEET

M14002 Data Sheet, Master Packaging

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 The Network D Circuit Board shall be manufactured in accordance with Drawing F8623286 and drawings pertaining thereto.

3.2 General specifications.- The circuit board shall meet the following requirements of MIL-F-13926, and MIL-M-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-M-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The circuit board shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the circuit board for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the circuit board shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The circuit board shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

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3.3.3 Temperatures.

3.3.3.1 Operating.- The circuit board shall meet the requirements of 3.5 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The circuit board shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating.- The circuit board shall meet the requirements of 3.5 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.3.4.2 Non-operating.- The circuit board shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The circuit board shall show no evidence of damage and shall meet the requirements of 3.4 after being inoculated with the composite spore suspension specified in MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The circuit board shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine spray of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Electrical performance.- The circuitry shall be in accordance with Electrical Schematic, Drawing F8623287. The symbols referenced in subsequent paragraphs pertain to this schematic and are used therein.

3.4.1 DUR-DLR driver.- The following conditions shall exist for each of the 3 DUL-DLR drivers.

3.4.1.1 Dynamic conditions - driver loaded.- With a symmetrical square wave varying from minus 0.2 plus or minus 0.2 volt to minus 6 plus or minus 1 volt applied to an input gate (figure 1a), and the load of figure 1b at the output, the output of the driver shall be a square wave swinging between the levels shown in figure 2. The input wave shall have the following properties: fundamental frequency of 10 ± 1 KC, with a rise and fall time of no more than 10 microseconds. The delay time between the input and output waves shall be as directed in figure 2.

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3.4.1.2 True state-static condition - (output loaded).- With a true condition applied to the input gate, and load of figure 1b, condition 1, at the output of the driver, the following conditions shall exist within a driver:

(a) The collector-emitter saturation voltage of the input transistor shall not exceed 0.22 volt.

(b) The input transistor base-to-emitter voltage shall not exceed 0.6 volt.

(c) With the output not loaded, the output transistor leakage current shall not exceed 15 microamps.

(c) The diode across the output transistor base-to-emitter junction shall develop a voltage no less than 0.1 volt but no greater than 0.5 volt.

3.4.1.3 False state-static condition - (output loaded).- With a false condition (-0.15 ± .15 volt) applied to the input gate, and the output loaded, the following conditions shall exist within a driver:

(a) The collector-emitter saturation voltage of the output transistor shall not exceed 0.22 volt.

(b) The emitter-to-base voltage of the output transistor shall not exceed 0.6 volt.

(c) The collector voltage of the output transistor shall be +3 ± 0.12 volts.

(d) The reverse leakage of the input transistor shall not exceed 15 microamps.

3.4.2 Regular driver.

3.4.2.1 The qualifications required of the DUR-DLR drivers (3.4.1) shall pertain to the regular driver with the exception:

(a) The simulated load shall be as directed in figure 1b, condition 2.

(b) The output voltage with a false signal applied to the input shall be -0.15 ± .15 volt.

3.4.3 Clock driver.

3.4.3.1 The qualifications required of the DUR-DLR drivers shall pertain to the clock driver with exception:

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(a) The simulated load shall be as directed in figure 1b, condition 3.

(b) The output voltage in the false state shall be +6 ± 1 volts.

3.4.4 Clock inhibit driver.

3.4.4.1 The dynamic requirements of 3.4.1.1 shall apply.

3.4.4.2 True state-static condition - (output loaded).- With a true condition (-6 ± 0.18 volts) applied to the input gate of figure 1a, and the load of figure 3 at the output, the following conditions shall exist within the driver:

(a) The voltage at the collector of the input transistor shall be -1.6 ± 0.15 volts.

(b) The voltage at the base of the output transistor shall be +3.3 ± 0.32 volts.

(c) With the output not loaded, the output transistor leakage current shall not exceed 15 microamps.

3.4.4.3 False state-static condition - (output loaded).- With a false condition (-0.15 ± 0.15 volt) applied to the input gate, and the output loaded, the following conditions shall exist within the driver:

(a) The voltage at the base of the output transistor shall be +2.69 ± 0.29 volts.

(b) The voltage across the zener diode shall be 6.2 ± 0.31 volts.

(c) The output transistor collector voltage shall be +2.9 ± 0.2 volts.

3.4.5 Neon driver.- The following requirements shall pertain to each of the 12 neon driver circuits.

3.4.5.1 True state - (output loaded).- With a true signal (-6 ± 0.18 volts) applied to the input gate, and the load of figure 4 connected to the output, the following conditions shall exist within a driver:

(a) The emitter voltage shall be -3.7 ± 0.22 volts.

(b) The collector voltage shall be +40 ± 10 volts.

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3.4.5.2 True state - (output not loaded).- With a true input to the driver, the voltage between the collector and the +150 volts shall be no less than 135 volts.

3.4.5.3 False state - (output not loaded).- With a false condition (-0.15 ±0.15 volt) applied to the input, the following conditions shall exist within a driver:

(a) The emitter voltage shall be no more negative than -1.1 volts.

(b) The collector voltage shall be +100 ± 10 volts.

(c) The requirement of condition (a) shall apply when any -6 volts, ground combination is used at the input gate.

3.4.6 Threshold amplifier.

3.4.6.1 Inhibit diode.

(a) The inhibit diode shall have a leakage current of no more than 0.1 microamps when reverse biased by 18 volts.

(b) The forward voltage drop across the diode shall not exceed 0.7 volt at 1.2 milliamps.

3.4.6.2 Static condition - (output and output prime loaded).- With separate loads identical to figure 5a connected to both the output and output prime of the amplifier, and the input network of figure 5b connected to the base of the input transistor, the following conditions shall exist within the amplifier:

(a) With the input adjusted to supply -3.92 ± 0.02 volts:

(1) The voltage at the output shall be between 0 and -0.25 volts.

(2) The voltage at the output prime shall be -6.8 ± 0.5 volts.

(b) With the input adjusted to supply -4.08 ± 0.02 volts:

(1) The voltage at the output shall be -6.8 ± 0.5 volts.

(2) The voltage at the output prime shall be between 0 volt and -0.25 volt.

3.4.7 Write switch conversion amplifier.- The following conditions shall exist for each of the four write switch conversion amplifiers.

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3.4.7.1 True state - (output loaded). - With the load of figure 6b connected to the output of the amplifier, and a true signal (ref. fig. 6a, switch closed) applied to the input, the following conditions shall exist within the amplifier.

(a) The base voltage shall be no less than +2.85 volts nor greater than +3.85 volts.

(b) With the output not loaded, the transistor leakage current shall not exceed 15 microamps.

3.4.7.2 False state - (output loaded). - With a false condition applied to the input (figure 6a, switch open), the following conditions shall exist within the amplifier:

(a) The base voltage shall be +2.6 ± 0.4 volts.

(b) The collector voltage shall be +2.85 ± 0.19 volts.

3.4.8 Resistor networks. - The resistors shall possess characteristics as specified on their appropriate drawings.

3.5 Temperature extremes. - The following requirements shall apply at the conditions of 3.3.3.1 and 3.3.4.1 (as applicable).

3.5.1 DUR-DLR driver, regular driver, clock driver (dynamic condition-driver loaded. - With a symmetrical square wave varying from -0.2 ± 0.2 volt to -6 ± 1 volt applied to an input gate (figure 1a), and the load of figure 1b at the output, the output of the driver shall be a square wave swinging between the levels of +3 ± 0.12 volt and -6 ± 1 volts (see figure 2). The input wave shall have the following properties: fundamental frequency of 10 ± 1 KC, with a rise and fall time of no more than 10 microseconds. The delay time between the input and output waves shall be as directed in figure 2.

3.5.2 Clock inhibit driver. - With the input of figure 1 and load of figure 3, the requirements of 3.4.1.1 shall apply.

3.5.3 Neon driver.

3.5.3.1 OFF state - (no load). - With a false input to the driver (any true-false combination input) the voltage at the collector with respect to the 150 volt supply shall not exceed 70 volts.

3.5.3.2 ON state - (no load). - With a true input to the driver, the voltage at the collector with respect to the 150 volt supply shall be no less than 110 volts.

3.5.3.3 ON state - (output loaded). - With the figure 4 load resistor at the output, the voltage across this resistor, with true inputs, shall be no less than 100 volts.

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3.5.4 Threshold amplifier.

(a) For the conditions of 3.4.6.2(a) the output voltage shall be between 0 volt and -0.5 volt. The output prime voltage shall be -6.8 ± 0.8 volts.

(b) For the conditions of 3.4.6.2(b) the output voltage shall be -6.8 ± 0.8 volts. The output prime voltage shall be between 0 volt and -0.5 volt.

3.5.5 Write switch conversion amplifier.- At the input and load conditions of figure 6, the following conditions shall exist within the amplifier:

(a) With a true condition (input switch closed) at the input of the amplifier, the collector voltage shall be -6 ± 0.42 volts.

(b) With a false input (input switch open), the collector voltage shall be +2.85 ± 0.25 volts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II, and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

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4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I
(100 percent inspection)

Characteristic	Requirement	Test Procedure
101. DUR-DLR driver dynamic operation	3.4.1.1	4.5.2.1.2
102. DUR-DLR driver true state-static	3.4.1.2	4.5.2.1.4
103. DUR-DLR driver false state-static	3.4.1.3	4.5.2.1.5
104. Regular driver	3.4.2	4.5.2.1
105. Clock driver	3.4.3	4.5.2.1
106. Clock inhibit driver-dynamic	3.4.4.1	4.5.2.2.1
107. Clock inhibit driver-true state-static	3.4.4.2	4.5.2.2.2
108. Clock inhibit driver-false state-static	3.4.4.3	4.5.2.2.3
109. Neon driver-true state (load)	3.4.5.1	4.5.2.3.2
110. Neon driver-true state (no load)	3.4.5.2	4.5.2.3.3
111. Neon driver-false state (no load)	3.4.5.3	4.5.2.3.4
112. Threshold amplifier-inhibit diode	3.4.6.1	4.5.2.4.1
113. Threshold amplifier static operation	3.4.6.2	4.5.2.4.3
114. Write switch conversion amplifier-true state	3.4.7.1	4.5.2.5.3
115. Write switch conversion amplifier-false state	3.4.7.2	4.5.2.5.4
116. Resistor network	3.4.8	4.5.2.6

4.3 Control tests.

4.3.1 General.- One circuit board shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings - visual
302. General specifications	3.2	MIL-F-13926 and MIL-C-46357-visual

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4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

	CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303.	Vibration	3.3.1	4.5.1.1
304.	Shock	3.3.2	4.5.1.2
305.	Humidity-Operating	3.3.4.1	4.5.1.3.2
306.	Humidity-Non-operating	3.3.4.2	4.5.1.3.1
307.	Salt spray	3.3.6	4.5.1.4
308.	Operating temperature	3.3.3.1	4.5.1.5
309.	Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

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Item Number		Description
1.	Power supply:	minus 18 ± 0.5 volts, current rating of at least 100 ma.
2.	Power supply:	minus 1.5 ± 0.04 volts, current rating of at least 30 ma.
3.	Power supply:	plus 15 ± 0.4 volts, current rating of at least 300 ma.
4.	Power supply:	plus 3 ± 0.12 volts, current rating of at least 30 ma.
5.	Power supply:	minus 6 ± 0.18 volts, current rating of at least 150 ma.
6.	Power supply:	plus 6 ± 0.12 volts, current rating of at least 100 ma.
7.	Power supply:	plus 35 ± 0.7 volts, current rating of at least 100 ma.
8.	Power supply:	minus 10 ± 0.2 volts, current rating of at least 100 ma.
9.	Power supply:	plus 150 ± 2 volts, current requirement of at least 200 ma. The peak-to-peak ripple and noise shall be less than 400 mv and the output impedance shall be less than 10 ohms from dc to 500 KC.
10.	Power supply:	minus 3 ± 0.12 volts, current requirement of at least 50 ma.

NOTE: The dc power supplies listed shall be variable to within 20 percent of the nominal voltage value, and have a regulation no worse than 2 percent. Unless otherwise specified, the peak-to-peak ripple shall be less than 100 mv, output impedance no greater than 2 ohms from dc to 500 KC. These supplies (items 1 through 10) shall deliver the voltages and tolerances as specified, for room ambient tests. Unless otherwise specified they shall be temperature compensated as shown below, and these values shall be used for all temperature extreme tests.

NOMINAL	MINUS 40°F	PLUS 135°F
-18 volts	-19.2 ± 0.3 volts	-16.7 ± 0.3 volts
-1.5 volts	-1.6 ± 0.03 volts	-1.4 ± 0.03 volts
+15 volts	+16.0 ± 0.3 volts	+14.9 ± 0.3 volts
+3 volts	+3.21 ± 0.06 volts	+2.79 ± 0.06 volts
-6 volts	-6.42 ± 0.12 volts	-5.58 ± 0.12 volts
+6 volts	+6.42 ± 0.12 volts	+5.58 ± 0.12 volts
+35 volts	+37.4 ± 0.7 volts	+32.5 ± 0.7 volts
-10 volts	-10.7 ± 0.2 volts	-8.3 ± 0.2 volts
+150 volts	+160 ± 3 volts	+140 ± 3 volts
-3 volts	-3.21 ± 0.06 volts	-2.79 ± 0.06 volts

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Item Number	Description
11. Generator, square wave:	Shall be capable of delivering a symmetrical square wave at a frequency of 250 kilo-cycles per second, and a voltage swing of between 0 volts and minus 6 volts. The rise and fall times of the output wave shall be no greater than 0.4 micro-seconds. The output impedance shall be no more than 100 ohms.
12. Oscilloscope:	Dual channel, rise time of scope plus measuring probe 50 millimicro-seconds or less, sweep period 50 millimicro-seconds per centimeter or greater, input resistance greater than 8 megohms, input capacitance less than 25 micro-micro-farads.
13. Miscellaneous resistors, diodes:	reference figures 1 thru 6.
14. Voltmeter, dc digital:	A range of at least .001 to 999 volts dc, or three significant figures. Accuracy of at least ± 0.2% of reading ± 1 count. Input impedance of at least 11 megohms on all ranges.
15. Ohmmeter:	any resistance measuring device with accuracies to at least 0.1 percent of the scale reading.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within ± 5 percent, and capable of measuring the frequency within ± 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the circuit board rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

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4.5.1.2 Shock.- This test shall be conducted in accordance with the conditions of 3.3.2 and Procedure II of the shock test defined in MIL-F-13926 except that impact shocks shall not exceed 5G's. The circuit board shall be subsequently inspected for physical damage or defects in compliance with 3.3.2. The unit shall then be inspected as specified in 4.5.1.3 prior to determining compliance with the requirements of 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 95 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.3 to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The circuit board shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2 for compliance with 3.3.6.

4.5.1.5 Temperature (operating and storage).- After being subjected to all environmental tests specified above, the circuit board shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.3 to determine compliance with the requirements of 3.5 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2 to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 Network D Circuit Boards produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

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4.5.2 Electrical performance.

4.5.2.1 Performance tests of DUR-DLR drivers, regulator driver, and clock driver.- The following tests shall determine compliance with 3.4.1, 3.4.2, and 3.4.3. Prior to energizing the module, separate input gates as shown in figure 1a shall be connected to pins 14B, 13B, 12B, 6A, and 7B.

4.5.2.1.1 Power supplies.- The power supplies listed in Table IV shall be connected.

TABLE IV

Power Supply	Pin #
plus 3 volts	16B
minus 1.5 volts	6B
minus 18 volts	4A
minus 6 volts	26A
plus 15 volts	26B
plus 6 volts	22B
power ground	17B

4.5.2.1.2 Dynamic tests - Test signal.- The standard input test signal to be used for all dynamic tests for the driver shall be an approximately symmetrical square wave (each half period shall be 50 ± 2.5 micro-seconds) varying from -0.2 ± 0.2 volt to -6 ± 1 volts with 10 percent to 90 percent rise and fall times each less than 10 microseconds. (Use item 11, par. 4.4.1.1). Apply the test signal through the input gate of figure 1a to the pins in column A, Table V in sequence. The load on each circuit will be connected as in figures 1b and 3. As the test signal is applied to a specific pin, the wave form observed at the corresponding pin in column B, Table V shall be a square wave, swinging between the levels indicated in column C, and at the frequency of the input test signal.

TABLE V

	Column A	Column B	Column C
DUR-DLR Drivers	14B 13B 12B	24A 23A 25B	+ 3 ± 0.12 volts to -6.2 ± 0.2 volts " "
Regular Driver	6A	24B	-0.15 ± .15 volt to -6.2 volts ± 0.2
Clock Driver	7B	23B	+ 6 ± 0.18 volts to -6.2 ± 0.2 volts
Clock Inhibit Driver	15B	25A	+ 3 ± 0.12 volts to -6.4 ± 0.2 volts

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4.5.2.1.3 Measurement of delay time between input and output waveforms.- Compare the signal applied to the input gate in column A, Table V to the output waveform at the corresponding pin in column B, on an oscilloscope. The delay time shall be in conformance with the requirements of figure 2.

4.5.2.1.4 True state-static condition (output loaded).

(a) Connect separate loads identical to figure 1b - condition 1 to pins 24A, 23A, and 25B. Connect the load of figure 1b - condition 2 to pin 24B. Connect the load of figure 1b, condition 3 to pin 23B.

(b) With minus 6 volts connected to the input gates, the voltage measured from pin 6B to the following points shall be no greater than 0.22 volts: TP14, TP16, TP18, TP20, and TP22.

(c) The voltage measured between pin 6B and the following pins shall be no greater than 0.6 volt: 14B, 13B, 12B, 6A and 7B.

(d) With the load removed the voltage measured at the following pins with respect to the minus 6 volt level shall be between zero volts and plus .016 volts: 24A, 23A, 25B, 24B and 23B.

(e) The voltage measured from column B to column A of Table VI shall be no less than 0.1 volt, but no greater than 0.5 volt.

TABLE VI

Column A	Column B
16B	TP15
16B	TP17
16B	TP19
17B	TP21
22B	TP23

4.5.2.1.5 False state-static condition (output loaded).

(a) Remove the minus 6 volt supply from each of the input gates and connect each of these gates to power ground.

(b) The voltage measured from column A to column B of Table VII shall be no greater than plus 0.22 volt.

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TABLE VII

Column A	Column B
16B	24A
16B	23A
16B	25B
17B	24B
22B	23B

(c) The voltage measured from column A to column B of Table VI shall be no greater than plus 0.6 volts.

(d) Disconnect the input gate from each of the pins in 4.5.2.1. The voltage measured between pin 17B and pins 14B, 13B, 12B, 6A and 7B shall not exceed 0.25 volts.

4.5.2.2 Clock inhibit driver test.- The following tests shall determine compliance with 3.4.4. Connect the input gate of figure 1a to pin 15B. Connect the load of figure 3 to pin 25A. Connect the power supplies of Table I to the appropriate pins and the dc returns of the power supplies to pin 17B.

4.5.2.2.1 Dynamic tests.

Repeat the applicable test of 4.5.2.1.2.

4.5.2.2.2 True state-static condition.

(a) Connect minus 6 ± 0.18 volts to the input gate.

(b) Measure the voltage at TP12. It shall be -1.6 ± 0.15 volts.

(c) The voltage measured at TP13 shall be + 3.3 ± 0.32 volts.

(d) With the load removed, the voltage measured at pin 25A with respect to the minus 6 volt level shall be between zero volts and + .016 volts.

4.5.2.2.3 False state-static conditions.

(a) Replace the minus 6 volts at the input gate with power ground.

(b) Measure the voltage at TP13. It shall be +2.7 ± 0.3 volts.

(c) The voltage measured from TP12 to TP13 shall be -6.2 ± 0.3 volts.

(d) The voltage at pin 25A shall be +2.9 ± 0.2 volts.

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4.5.2.3 Neon driver tests.- The following tests shall determine compliance with 3.4.5.

4.5.2.3.1 Power and load connections.

(a) Connect the power supplies to Table VIII to their corresponding pins. Connect separate loads, identical to figure 4, to the pins of column C, Table VI.

TABLE VIII

Power Supplies for Neon Driver

Power Supply	Pin #
Minus 3 volts	4B, 20B, 29B, 60B
Minus 18 volts	11A, 32A, 56A
Plus 150 volts	2A, 28A, 64A

(b) The power supplies listed in Table VIII shall have returns jumpered together and connected to pins 1B, 27B, and 65B.

4.5.2.3.2 True state - (output loaded).

(a) Apply a true signal (-6 ± 0.18 volts) to each of the pins of column A, Table IX. The voltage at each of the test points of column B shall be -3.7 ± 0.22 volts.

(b) The voltage between the pins of column C, Table IX, and ground shall be 40 ± 10 volts.

4.5.2.3.3 True state (output not loaded).

(a) Apply a true signal (-6 ± 0.18 volts) to each of the pins of column A, Table IX. The voltage at the pins of column C with respect to the + 150 volts shall be no less than 135 volts.

4.5.2.3.4 False state (output not loaded).

(a) Remove the minus 6 volts from column A2, Table IX, and replace it with power ground. The voltage at each of the test points of column B, shall be no more negative than -1.1 volts.

(b) The voltage between the pins of column C, and ground shall be + 100 ± 10 volts.

(c) Interchange the voltage levels specified in (a), and make the measurements required of that paragraph.

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TABLE IX

Column A 1	Column A 2	Column B	Column C
7A,	8B	TP24	2B
8A,	9B	TP25	3B
9A,	11B	TP26	5B
17A,	16A	TP27	22A
15A,	14A	TP28	21A
13A,	12A	TP29	20A
31B,	32B	TP30	33A
30A,	31A	TP31	30B
60A,	59B	TP32	64B
59A,	58B	TP33	63B
58A,	57B	TP34	62B
57A,	56B	TP35	61B

4.5.2.4 Threshold amplifier tests.- The following tests shall determine compliance with 3.4.6.

4.5.2.4.1 Inhibit diodes.

(a) Apply a minus 18 volt supply through a 0.5 megohm ± 5 percent resistor to pin 54A. Ground pin 54B. The voltage across the .5 megohm resistor shall be less than 0.1 volts.

(b) Remove the minus 18 volt supply and resistor connected to pin 54A. Apply a + 15 volt supply through a 12K ohm ± one percent resistor to pin 54A. Ground pin 54B. The voltage at pin 54A with respect to ground shall not exceed 0.7 volts.

4.5.2.4.2 Power and ground connections.

(a) Connect the following power supplies:

Plus 35 volts to pin 51A.
Plus 6 volts to pin 61A.
Minus 10 volts to pin 53B.

(b) Connect the dc returns of these supplies to pin 55A.

4.5.2.4.3 Static conditions.

(a) Separate loads, identical to figure 5a, shall be connected to pins 55B and 62A. Connect the input network of figure 5b to pin 54B.

(b) With a voltmeter connected from 54B to ground, the input pot shall be adjusted until the voltmeter measure -4.0 ± 0.02 volts. Adjust the potentiometer R30 until the voltage at 55B teeters between minus 7 and 0 volts.

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(c) Adjust the input network to supply -3.92 ± 0.02 volts to 54B.

(1) The voltage at 62A shall be between 0 volt and minus 0.25 volt.

(2) The voltage at 55B shall be -6.8 ± 0.5 volts.

(d) Adjust the input network to supply -4.08 ± 0.02 volts.

(1) The voltage at 62A shall be -6.8 ± 0.5 volts.

(2) The voltage at 55B shall be between 0 volt and minus 0.25 volt.

4.5.2.5 Write switch conversion amplifier tests.- The following tests shall determine compliance with 3.4.7.

4.5.2.5.1 Power and ground connections.

(a) Connect the minus 6 volt supply to pin 51B, and the plus 3 volt supply to pin 16B.

(b) The dc returns of the power supplies shall be connected to pin 55A.

4.5.2.5.2 Input and load connections.

(a) Connect the input of figure 6a to pins 41A, 40B, 40A, and 39B.

(b) Loads, as per figure 6b, shall be connected to pins 48A, 49B, 47A, and 48B.

4.5.2.5.3 True state.

(a) Close the input switch of figure 6a.

(b) The voltage at TP1, TP2, TP3, and TP4, with respect to ground, shall be no less than + 2.85 volts, but not in excess of + 3.85 volts.

(c) With the load removed, the voltage measured at pins 48A, 49B, 47A, and 48B with respect to the minus 6 volt level shall be between zero volts and plus 0.33 volts.

4.5.2.5.4 False state.

(a) Open the input switch of figure 6a.

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(b) The voltages measured at TP1, TP2, TP3, and TP4, shall be + 2.6 ± 0.4 volts.

(c) The voltage at pins 48A, 49B, 47A, and 48B, shall be + 2.85 ± 0.2 volts.

4.5.2.6 Resistor network tests.- The following tests shall determine compliance with 3.4.8.

(a) An ohmmeter (item 15; 4.4.1.1) shall be connected to pin 42A and a measurement made at each of pins 49A, 34B, 35A, 33B, 34A, 35B, 36A, 36B, 37A, 37B, 38A, 38B, and 39A. The resistance shall be 27000 ohms ± 1 percent.

(b) Connect the ohmmeter to pins 52B and 46B. The resistance shall be 600 ohms ± 5 percent. Measure from 52B to 45A. The resistance shall be 1000 ohms ± 5 percent.

4.5.3 Temperature extremes.- The following tests shall be performed under the test conditions of 4.5.1.3.2 and 4.5.1.5 using the temperature compensated supply voltages of 4.4.1.1. These tests shall determine compliance with 3.5.

4.5.3.1 DUR-DLR drivers, regular driver, clock driver.

Repeat as in 4.5.2.1.2 except the output rise and fall times shall not exceed 20 micro-seconds.

4.5.3.2 Clock inhibit driver.

Repeat as in 4.5.2.1.2 except the output rise and fall times shall not exceed 20 micro-seconds.

4.5.3.3 Neon driver.

4.5.3.3.1 OFF State - (no load).

(a) Make the supply connections listed in column A of Table X.

(b) Apply minus 6 volts to the pins of column A1, Table IX. Ground the pins in column A2.

(c) The voltage between the plus 150 volts and the pins of column C shall not exceed 70 volts.

(d) Reverse the input connections of (b) and perform the test in (c).

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4.5.3.3.2 ON state - (no load).

(a) Make the supply connections of column B, Table X.

(b) Apply minus 6 volts to all the pins of column A, Table IX.

(c) The voltage between the plus 150 volts and the pins of column C, Table IX, shall be no less than 110 volts.

4.5.3.3.3 ON state - (output loaded).

(a) Repeat (a) and (b) of 4.5.3.3.2.

(b) Connect figure 4 to the pins of column C, Table IX.

(c) The voltage between the plus 150 volts and the pins of column C shall be no less than 100 volts.

TABLE X

Voltage		Connect to pins
Column A	Column B	
+160 ± 3 volts	+140 ± 3 volts	2A, 28A, 64A
-19.2 ± 0.3 volts	-16.7 ± 0.3 volts	11A, 32A, 56A
-3.15 ± 0.06 volts	-2.79 ± 0.06 volts	4B, 20B, 29B, 60B
POWER GROUND	POWER GROUND	1B, 27B, 65B

4.5.3.4 Threshold amplifier.- Repeat 4.5.2.4 with the following changes: The voltage measured in 4.5.2.4.1(a) shall be less than 0.25 volts, the voltage measured in (b) shall not exceed 1 volt. The false outputs of the circuit shall not exceed 0.5 volts and the true output shall be 6.8 ± 0.8 volt.

4.5.3.5 Write switch conversion amplifier.

(a) With the conditions and external connections of 4.5.2.5.1 and 4.5.2.5.2, and a true input, the voltages at pins 48A, 49B, 47A, and 48B, shall be -6 ± 0.5 volts.

(b) With a false input, the voltages at pins 48A, 49B, 47A, and 48B, shall be + 2.85 ± 0.25 volts.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet, Master No. 14002.

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6. NOTES

6.1 Intended use.- The eight functional circuits contained on Network D, with their intended use, are as follows.

6.1.1 Write switch conversion amplifier.- The write switch conversion amplifier provides current amplification to crosspoints and for matching of write switches.

6.1.2 DUR-DLR drivers.- Provide amplified logic gage inputs to activate particular crosspoints of the matrix.

6.1.3 Regular driver.- The regular driver provides power amplification for driving the logic gates.

6.1.4 Clock driver.- The clock driver provides amplification of the logic output of primary gates to supply clock triggering for setting of the flip-flops.

6.1.5 Neon driver.- The neon driver provides amplification of primary gate outputs to drive associated neon indicators.

6.1.6 Threshold amplifier.- The threshold amplifier is used to detect whether or not signal level at input is above predetermined reference level.

6.1.7 Resistor network.- The resistor network provides; biasing voltages for OR driving capabilities, OR gate functions for driving primaries of nixie drivers, and for use with the TFI, TFO and TAP test circuits.

6.1.8 Clock inhibit driver.- Provides power gain for the ISR and OSR drivers.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet (see 5.1).
(e) A block of serial numbers to identify each circuit board.

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

Custodian:

Army - MU

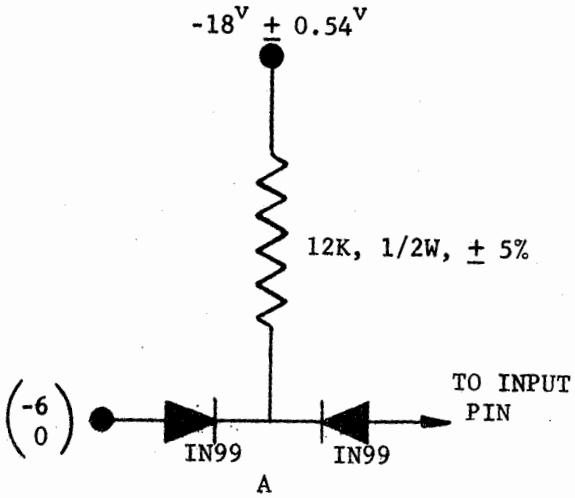
Preparing activity:

Army - MU

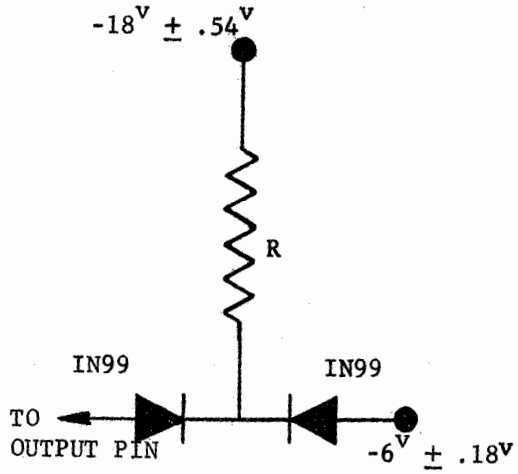
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NETWORK "D"



A



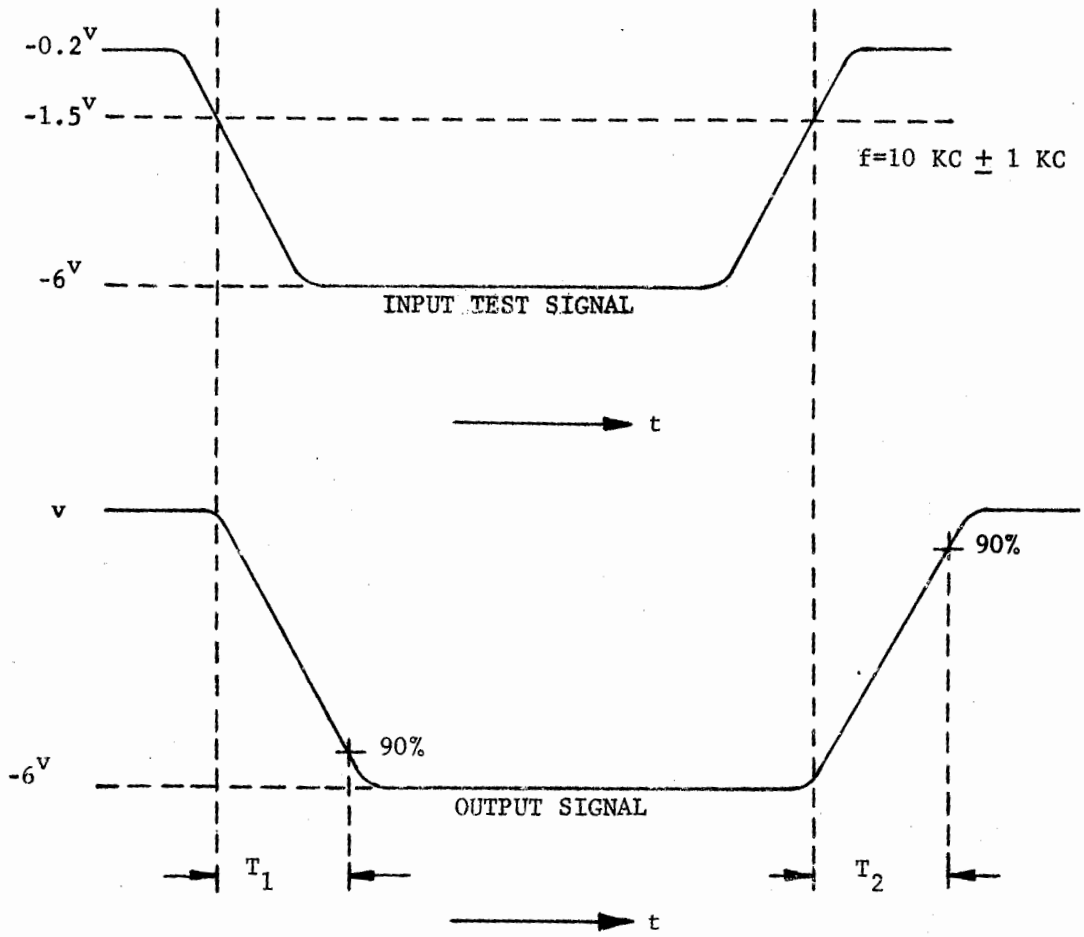
DUR-DLR	CONDITION 1	R=1.2K Ω, 1W, 5%
REGULAR	CONDITION 2	R=470 Ω, 1W, 5%
CLOCK	CONDITION 3	R=2K Ω, 1W, 5%

B

INPUT GATE & LOAD FOR DRIVERS

FIGURE 1

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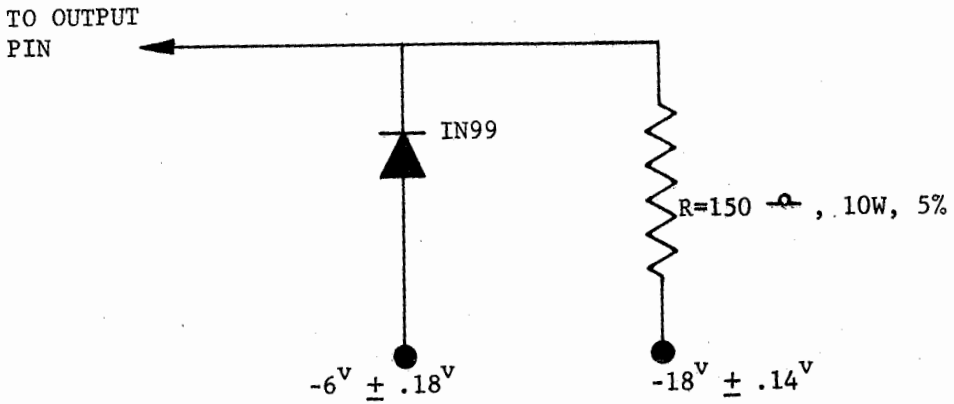


Circuit	Output V	T1 less than:	T2 less than:
DUR-DLR	+3 ± 0.12	400 nano sec.	400 nano sec.
Regular	-0.15 ±0.15	"	"
Clock	+6 ± 1	500 nano sec.	500 nano sec.
Clock inhibit	+3 ± 0.12	"	"

Delay time Comparison of Input and Output waveforms. The delay time (T1) measured from the time the square wave applied to the input gate reaches minus 1.5 volts (going towards minus 6 volts) until the output reaches 90% of its total excursion (going towards minus 6 volts) and delay time measured from the time the square wave applied to the input gate reaches minus 1.5 volts (going towards zero volts) until the output reaches 90% of its total excursion (going towards the false state plus 3 volts) shall be as indicated in the table. For extreme temperatures add 5% to the limits of T1 and T2.

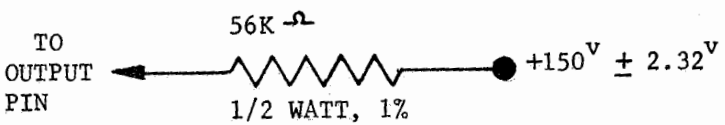
FIGURE 2

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LOAD FOR CLOCK INHIBIT DRIVER

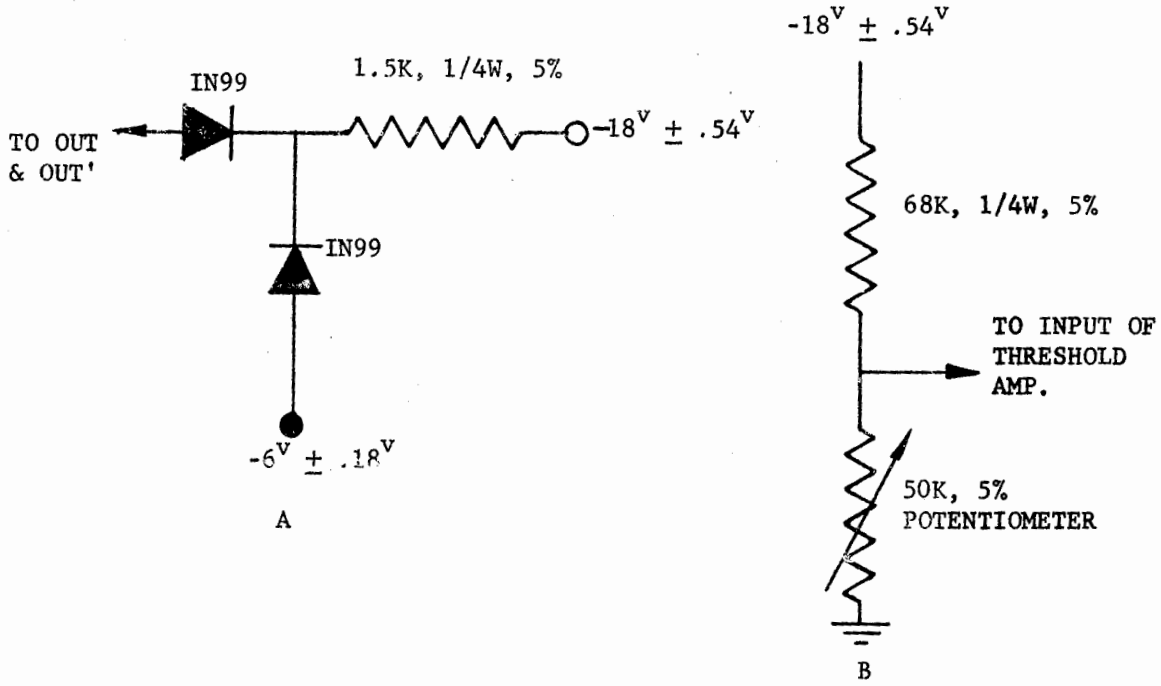
FIGURE 3



NEON DRIVER LOAD

FIGURE 4

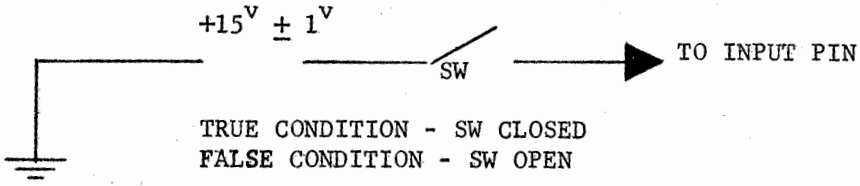
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INPUT AND LOAD FOR THRESHOLD AMPLIFIER

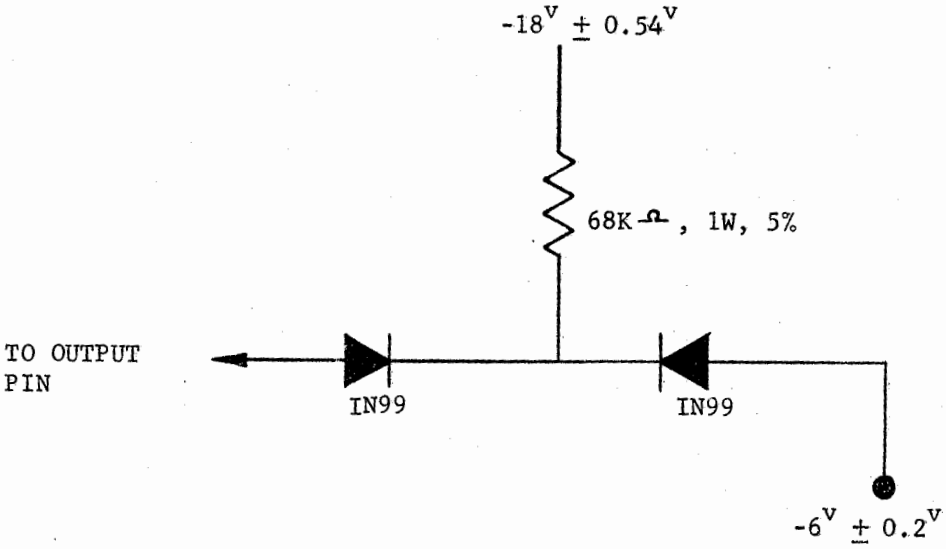
FIGURE 5

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INPUT NETWORK FOR WRITE SW AMP

A



LOAD FOR WRITE SW AMP

B

FIGURE 6

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